

Standard Operating Procedure

LabVIEW Continuous/Periodic Camera Acquisition and Automatic PNG Saving VI

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Application: Continuous camera acquisition and automatic sequential PNG saving from a selected camera

Document title	LabVIEW Continuous Camera Acquisition and Automatic PNG Saving VI SOP	Version	1.0
Prepared for	Operating users / lab personnel	Prepared by	Digambar Killedar
Output format	Sequential PNG image files	Primary function	Acquire, display, and save image frames continuously until STOP is pressed

1. Purpose

This SOP describes the standard method for operating the LabVIEW continuous camera acquisition VI shown in the attached front panel and block diagram. The VI opens a selected camera session, displays the live image, acquires frames continuously at the user-defined frame rate, and writes each frame automatically to disk in PNG format until the STOP button is pressed.

2. Scope

This procedure is intended for laboratory users, operators, and students who need to capture image sequences from the configured camera system for inspection, documentation, or later image processing. It applies to routine continuous image acquisition using NI Vision / IMAQdx with automatic numbered file saving.

3. Pre-startup checks

- Confirm that the required camera is physically connected and visible in NI MAX or the Vision acquisition system.
- Confirm that the intended output folder configured in the block diagram exists and that the user has write permission to that folder.
- Verify that the selected Frame rate is greater than zero so that a valid loop delay can be calculated.
- Check that the chosen Image Type is appropriate for the intended acquisition mode.
- Confirm that enough disk space is available because the VI saves frames continuously until it is stopped.

4. Front panel controls and indicators

Item	Function	Operator guidance
STOP	Stops the acquisition loop and ends automatic file saving.	Press only after the required number of images has been collected.
Image Name	Defines the name used when the NI Vision image buffer is created.	Use a valid descriptive name and keep it consistent with the selected image type.
Session In	Selects the camera source used for acquisition.	Choose the correct connected session, for example cam6.

Item	Function	Operator guidance
Frame rate	Controls the acquisition loop timing using $1 / \text{Frame rate} \times 1000$ ms.	Use a positive value; lower values reduce the saving rate and file count.
Frames	Displays the running frame count.	Use this indicator to confirm that acquisition is progressing.
Connected Only? (Yes)	Filters the camera information query to connected devices only.	Keep enabled for routine operation unless disconnected devices must be listed.
Image Type	Sets the image type used when creating the IMAQ image buffer.	Select the required format, for example Grayscale (U8), before starting the VI.
Camera Information Array	Displays camera details such as interface name, vendor, model, and serial data.	Use this array to verify that the correct camera has been opened.
Image display	Shows the current acquired image from the selected camera.	Use this window to confirm focus, field of view, gain and exposure(to be included) during acquisition.

5. Operating procedure

- 1. Open the VI:** Launch the LabVIEW VI and verify that the front panel loads without errors.
- 2. Select the camera:** From Session In, choose the required camera, for example cam6.
- 3. Set the image buffer parameters:** Enter the desired Image Name and confirm the Image Type before acquisition begins.
- 4. Review the camera information:** Leave Connected Only? enabled and confirm in the Camera Information Array that the correct FLIR camera is identified.
- 5. Set the frame rate:** Enter the required Frame rate. For example, a value of 2 targets approximately one saved frame every 0.5 seconds.
- 6. Confirm the output folder:** Review the fixed save path configured in the block diagram and ensure that the folder exists before running the VI.
- 7. Run the VI:** Click Run in LabVIEW. The Image display should populate with the live image from the selected camera.
- 8. Monitor acquisition:** Observe the Frames indicator increasing and confirm that PNG files are being created sequentially in the configured folder.
- 9. Stop the VI:** Press STOP after the required capture period is complete. The acquisition loop ends and saving stops.
- 10. Verify the saved sequence:** Open the destination folder in Windows Explorer and confirm that the numbered PNG files were created successfully.

Important: In the current implementation, the save folder is defined inside the block diagram rather than on the front panel. If the path is changed in code, verify it again before running the VI.

6. Troubleshooting

Issue	Likely cause	Recommended action
No image appears in the display	Camera is not connected, not recognized, or the wrong session was selected.	Check hardware connection, verify the camera in NI MAX, and reselect the correct Session In device.
Files are not being saved	Output folder does not exist, the path is invalid, or no permission to write.	Create the folder first, verify the configured path in the block diagram, and save to a writable location.
Saving rate is too fast or too slow	Frame rate value is incorrect or zero.	Enter the intended positive frame-rate value

Issue	Likely cause	Recommended action
		and rerun the VI.
Camera Information Array is blank or incomplete	Connected Only? setting, session selection, or camera query failed.	Verify the selected camera and check the connected-only setting before running again.
Unexpected image format	Incorrect Image Type was chosen when the image buffer was created.	Stop the VI, select the correct Image Type, and restart acquisition.

7. Functional summary of the VI

- The block diagram initializes the selected camera session and creates an NI Vision image buffer using the specified Image Name and Image Type.
- The VI runs inside a loop so the operator can view the live image while frames are acquired and saved automatically.
- The Frame rate control is converted to a loop delay using $1 / \text{Frame rate} \times 1000 \text{ ms}$, which governs the approximate saving interval.
- Each loop iteration displays the current frame and writes a PNG file using the configured base path and the running frame number.
- The Frames indicator shows the current frame count, and the STOP control exits the loop when acquisition is complete.
- The Camera Information Array provides quick verification of the connected device details during setup.

Appendix A. Reference screenshots

A1. Front panel

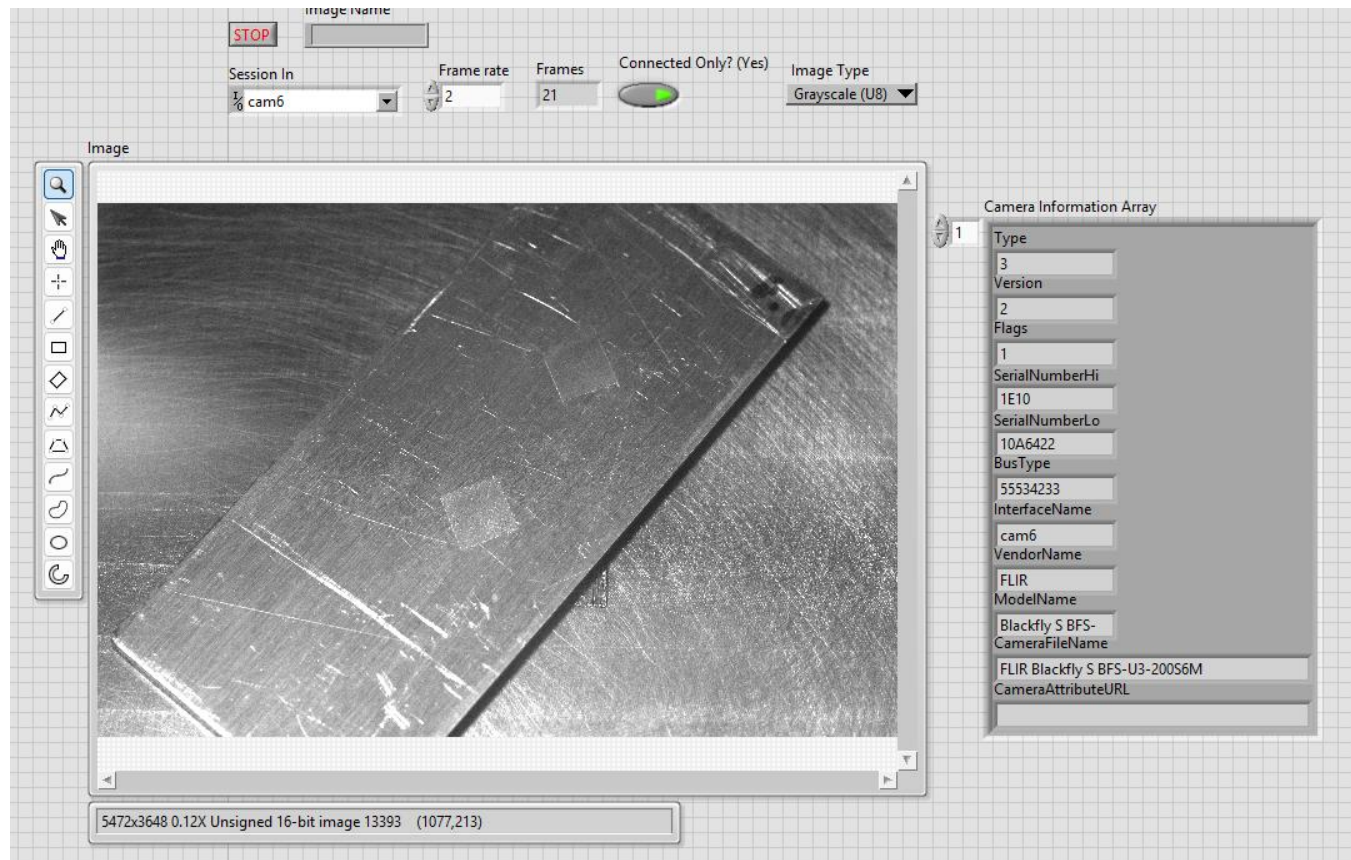


Figure A1. Front panel used for camera selection, image preview, frame counting, camera information display, and continuous image acquisition.

A2. Block diagram

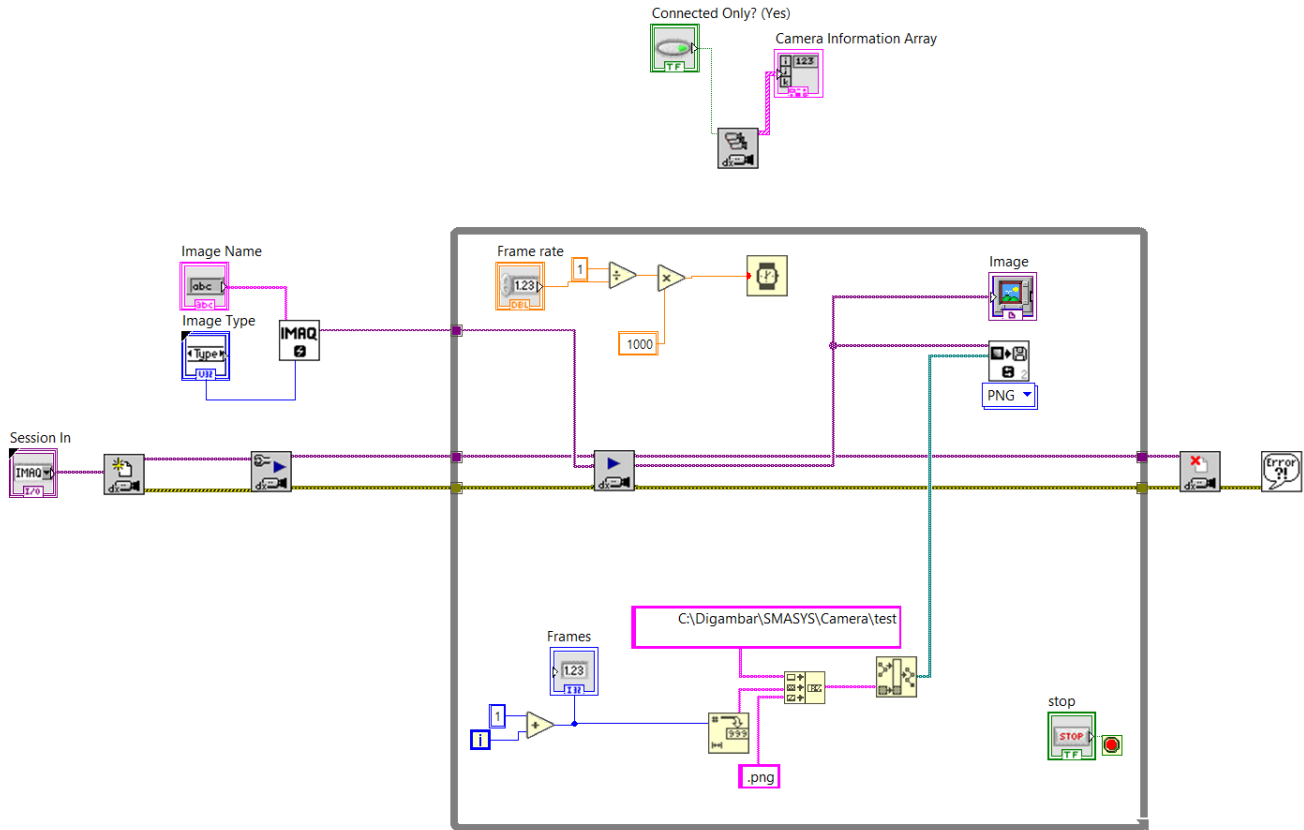


Figure A2. Block diagram showing session initialization, image buffer creation, frame-rate timing, live image display, automatic PNG writing, frame numbering, and STOP logic.